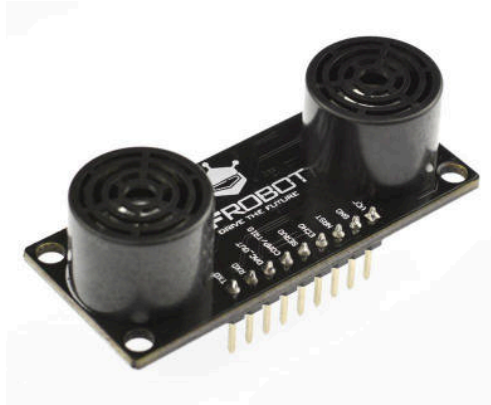
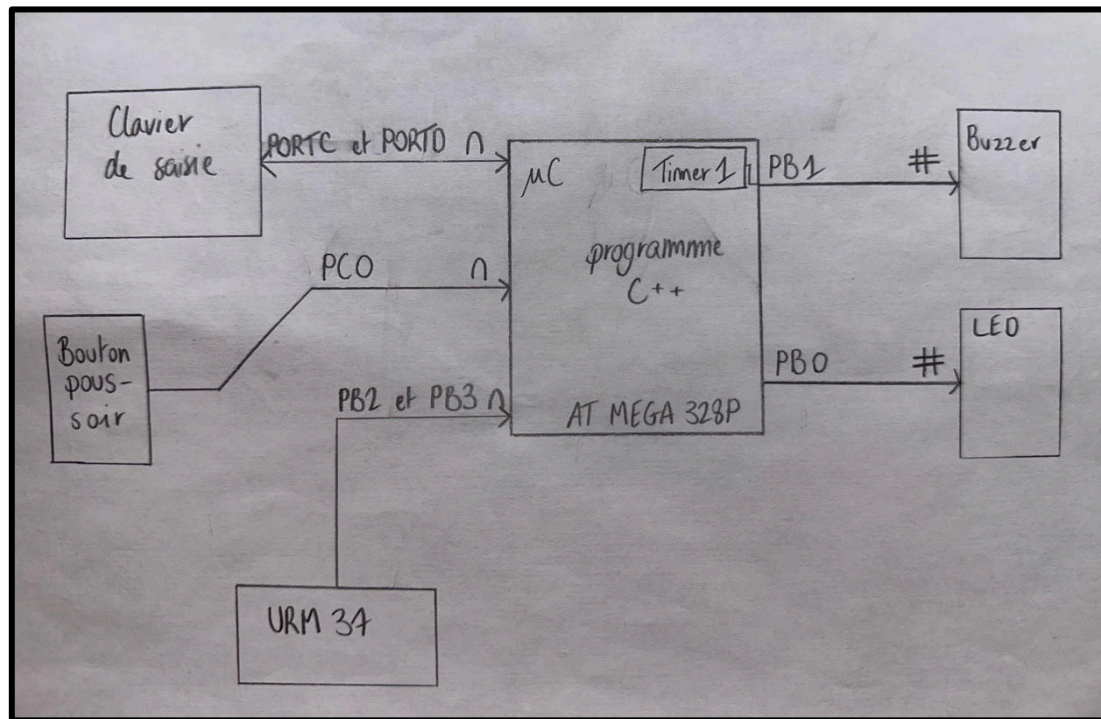


Projet SAé : Alarme Ultra Son





Synoptique :



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2.

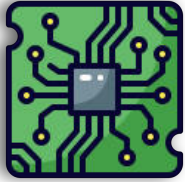
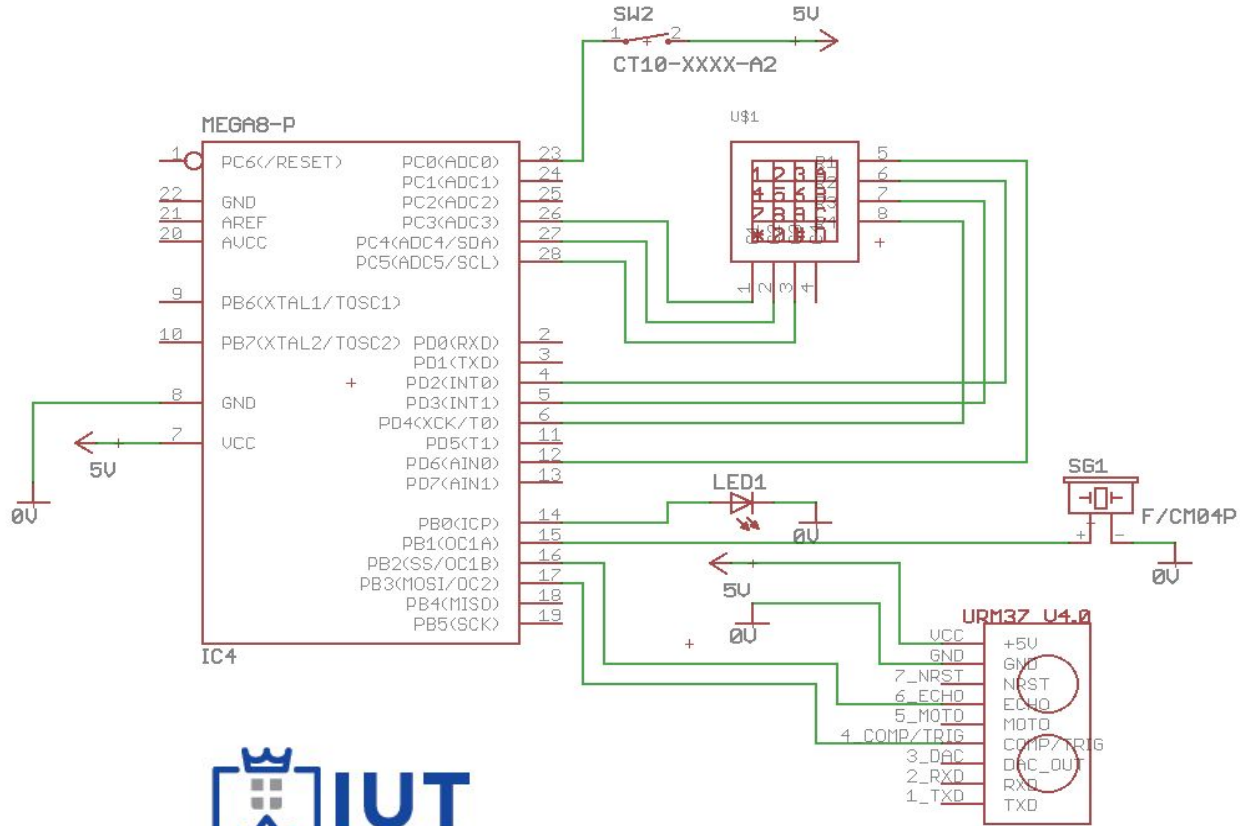


schéma structurel :



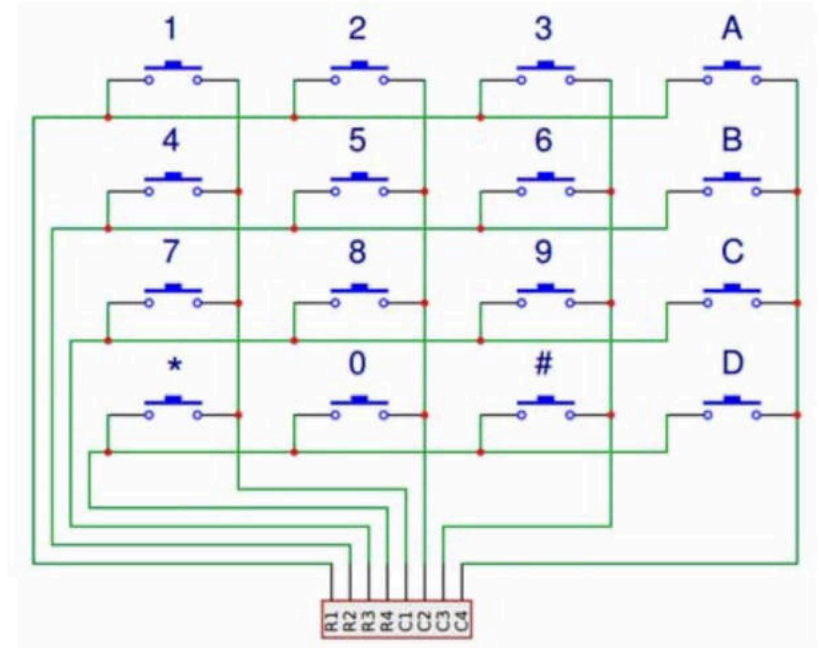
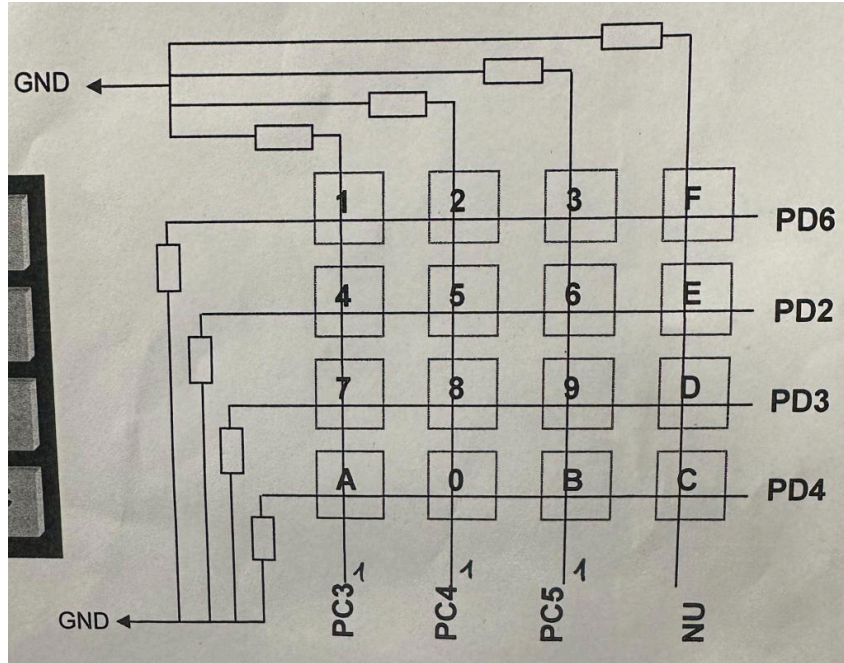
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3.

Zoom technique, clavier matriciel :




```
//on retrouve la valeur du clavier cor  
val = tab[line_value][column_value];  
  
return val;
```


Zoom technique, le timer :



Utilisation du TIMER 1 pour l'alarme

- signal alternatif

Configuration des registres :

$$f_{OCnxPFCPWM} = \frac{f_{clk\ I/O}}{2 \cdot N \cdot TOP}$$

Mode PWM

The N variable represents the prescaler divider (1, 8, 64, 256, or 1024).

Intérêt du mode PWM

- Fournir une tension carré
- Moduler le rapport cyclique

$$ICR1 = TOP = \frac{f(clk)}{2 \cdot N \cdot f}$$

=> Valeur **maximum** de comptage du Timer

Zoom technique, le timer :



TCCR1A – Timer/Counter1 Control Register A

Bit	7	6	5	4	3	2	1	0	
(0x80)	COM1A1	COM1A0	COM1B1	COM1B0	–	–	WGM11	WGM10	TCCR1A
Read/Write	R/W	R/W	R/W	R/W	R	R	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

TCCR1B – Timer/Counter1 Control Register B

Bit	7	6	5	4	3	2	1	0	
(0x81)	ICNC1	ICES1	–	WGM13	WGM12	CS12	CS11	CS10	TCCR1B
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 15-3. Compare Output Mode, Phase Correct and Phase and Frequency Correct PWM⁽¹⁾

COM1A1/COM1B1	COM1A0/COM1B0	Description
0	0	Normal port operation, OC1A/OC1B disconnected.
0	1	WGM13:0 = 9 or 11: Toggle OC1A on Compare Match, OC1B disconnected (normal port operation). For all other WGM1 settings, normal port operation, OC1A/OC1B disconnected.
1	0	Clear OC1A/OC1B on Compare Match when up-counting. Set OC1A/OC1B on Compare Match when downcounting.
1	1	Set OC1A/OC1B on Compare Match when up-counting. Clear OC1A/OC1B on Compare Match when downcounting.

Table 15-5. Clock Select Bit Description

CS12	CS11	CS10	Description
0	0	0	No clock source (Timer/Counter stopped).
0	0	1	$clk_{IO}/1$ (No prescaling)
0	1	0	$clk_{IO}/8$ (From prescaler)
0	1	1	$clk_{IO}/64$ (From prescaler)
1	0	0	$clk_{IO}/256$ (From prescaler)
1	0	1	$clk_{IO}/1024$ (From prescaler)
1	1	0	External clock source on T1 pin. Clock on falling edge.
1	1	1	External clock source on T1 pin. Clock on rising edge.

Zoom technique, le timer :



Table 15-4. Waveform Generation Mode Bit Description⁽¹⁾

Mode	WGM13	WGM12 (CTC1)	WGM11 (PWM11)	WGM10 (PWM10)	Timer/Counter Mode of Operation	TOP	Update of OCR1x at	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	BOTTOM	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	BOTTOM	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	BOTTOM	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	(Reserved)	–	–	–
14	1	1	1	0	Fast PWM	ICR1	BOTTOM	TOP
15	1	1	1	1	Fast PWM	OCR1A	BOTTOM	TOP

Note: 1. The CTC1 and PWM11:0 bit definition names are obsolete. Use the WGM12:0 definitions. However, the functionality and location of these bits are compatible with previous versions of the timer.

Zoom technique, le timer :



Les bits ICNC1 et ICES1 seront positionnés à 0

prescaler = 1

PWM Phase and Frequency Correct avec val max comptage ICR1

On obtient :

TCCR1A = 0x80; //0b10000000

TCCR1B = 0x11; //0b00010001

OCR1A = 0.5 * ICR1; // 50 % (rapport cyclique)

Activer / Désactiver le Timer

TCCR1B = 0x00; //0b00000000 / TCCR1B
= 0x11; //0b00010001

câbler le buzzer sur la sortie PB1 (=Output
Compare Timer1)

Faire varier ICR1 pour faire varier la
fréquence du buzzer

Zoom technique, le timer :



```
void setup() {
```

```
  TCCR1A = 0x80; //0b10000000
```

```
  TCCR1B = 0x11; //0b00010001
```

```
  ICR1 = 8000000 / (2 * 1 * 500); // TOP (Fclk/(2*prescaler*frequence))
```

```
  OCR1A = 0.5 * ICR1;           // 50 % (rapport cyclique)
```

```
  DDRB = 0x02; //0b00000010, place le port PB1 en sortie
```

```
}
```

```
uint16_t freq = 500;
```

```
bool sens = 1; //si sens = 1, on incrémente la fréquence, sinon, on décrémente le timer
```

```
void loop() {
```

```
  if (sens) {
```

```
    freq -= 3; //décrémenter le timer
```

```
  } else {
```

```
    freq += 3; //incrémenter le timer
```

```
  }
```

```
  if (freq > 1000) { //On inverse le sens de l'incrémenter le timer en fonction de sa valeur de la fréquence
```

```
    sens = !sens;
```

```
  } else if (freq < 500) {
```

```
    sens = !sens;
```

```
  }
```

```
  ICR1 = 8000000 / (2 * 1 * freq); // modification de la fréquence
```

```
  OCR1A = 0.5 * ICR1;           // 50 %
```

```
}
```

Bilan :



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11.